

DAC Testnet Inception Debugging Report

Frontend—Blockchain State Desynchronization, RPC Reliability, and Reward Accounting Validation

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Target	DAC Inception Testnet Web v0.9.7 — https://inception.dachain.io
Environment	WSL Ubuntu local DAC node · Herond Browser · Indonesia / CGNAT
Report Type	Application-Layer Consistency, Infrastructure, and Reward Accounting Investigation
Relates to	Reports 1–8 of the DAC Testnet (Inception) Technical Report Series
Report Archive	https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup/tree/main/Testnet_(Inception)_Technical_Reports

Executive Positioning: This report consolidates checkpoints v0.1–v0.7 into a single professional technical submission. The investigation focuses on the consistency boundary between DAC Inception frontend state, backend/indexer state, public RPC state, wallet/local RPC state, explorer state, and on-chain contract execution.

Executive Summary

This report consolidates a full-web debugging investigation of DAC Inception Testnet Web v0.9.7. The work began with faucet and RPC inconsistencies, then expanded across My Activity, Leaderboard, Settings and Social Binding, Quantum Crate, QE Pool/Exchange, and Badges/Rank NFT state.

The primary conclusion is that the DAC blockchain and several smart contract flows generally execute correctly on-chain, but the application layer can fail to reconcile confirmed on-chain state into the frontend dashboard, backend profile, indexer history, or reward accounting systems.

Core Finding: Across multiple modules, the same failure pattern appears — on-chain state is confirmed through local RPC and/or explorer, while frontend/backend state remains stale, incomplete, or misleading. This affects balances, profile data, faucet state, leaderboard metadata, QE reward crediting, and NFT mint status.

High-Level Findings

- Public RPC lag and instability — including stale block heights, null receipts for locally confirmed transactions, 502 Bad Gateway, 504 Gateway Time-out, and connection reset errors.
- DAC Inception UI components can continue displaying stale state while presenting a live experience to users.
- Faucet cooldown can start after request acceptance even when delivery remains pending and no transaction hash is available.
- My Activity mixes state sources: profile-derived TX count/QE data and public-RPC-derived DACC balance can disagree on the same page.
- Leaderboard search, sorting, and pagination behave as client-side operations over a limited preloaded dataset despite presenting a global ranking experience.
- Settings confirms basic social-linked booleans but does not surface role-level Discord eligibility, early badge tier logic, or rebind/recovery actions.

- Quantum Crate QE accounting worked in the tested session, but immediate UI/history refresh lag and DACC drop settlement opacity remain issues.
- QE Pool stake/unstake and burn contract mechanics worked on-chain, but frontend confirmation stuck and a confirmed burn did not credit the expected QE within the observation window.
- Rank NFTs for Vanguard and Sentinel were confirmed on-chain but still appeared as mintable because `profile.nft_tx_hash` was missing.

Balanced Assessment

The investigation does not indicate that every subsystem is broken. Several contract paths and backend calculations worked as expected: Quantum Crate QE accounting was correct after refresh, QE Pool stake/unstake mechanics were valid, and already-indexed burn history aligned with the stated 1 DACC = 1000 QE conversion model. The concern is specifically the reliability of state propagation and reconciliation after user actions.

Scope and Methodology

The investigation was structured as a sequence of checkpoints from v0.1 to v0.7. Each checkpoint focused on a module or state boundary and was validated through a combination of UI observation, Network tab inspection, browser Console calls, local/public RPC comparison, explorer verification, and controlled on-chain transactions where appropriate.

Checkpoint	Module Focus	Primary Validation Method
v0.1	Clean environment, faucet, public RPC, profile, balance display	Incognito observation, API inspection, local/public RPC comparison
v0.2	My Activity	profile/, sync/, qe-history/, public RPC comparison
v0.3	Leaderboard	Leaderboard endpoint, search/sort/pagination observation
v0.4	Settings, social binding, QE Pool baseline, faucet finalized flow	profile/, queue/status endpoints, auth/session observations
v0.5	Quantum Crate	open/, history/, profile/, UI state comparison
v0.6	QE Pool / Exchange active flow	Stake, unstake, burn, local/public RPC receipts, exchange history, recovery
v0.7	Badges / Rank NFT state	profile.badges, UI, explorer inventory, local RPC eth_getLogs

Evidence Standard

Findings are separated into confirmed observations, interpretations, and unconfirmed hypotheses. Confirmed findings require at least one concrete source — API response, RPC response, explorer result, transaction receipt, raw trace, or deterministic UI behavior. When a claim is based on user experience without real-time debug proof, it is marked as a UX observation rather than a confirmed on-chain failure.

Limitations

- This investigation is based on one primary tester wallet. Some findings may require wider user sampling.

- Public RPC instability affected some browser-console checks. Local RPC and explorer were used as stronger sources where public RPC was stale or unavailable.
- One faucet observation sequence was interrupted by a device restart caused by a Windows update. Post-restart state was still useful as cooldown and missing-delivery state persisted after relogin.
- Some endpoints returned HTML app shell responses when guessed routes did not map to JSON endpoints; these are treated as route/observability limitations, not endpoint failures unless confirmed by the app itself.

Environment and Tooling

Web application	DAC Inception Testnet Web v0.9.7
Web app URL	https://inception.dachain.io
Explorer	https://exptest.dachain.tech
Public RPC	https://rpctest.dachain.tech
Local RPC	http://127.0.0.1:8546
Primary wallet	0x870ad63acc507cdfd878f170606d19ae78988afe
Browser	Herond (Chromium-based)
Node environment	WSL Ubuntu local DAC node
Network context	Indonesia / residential CGNAT environment

Wallet Environment Note

Incognito-mode observations used OKX Wallet through WalletConnect. This was separate from the OKX browser extension environment. Active on-chain transaction testing was performed in the normal browser session because that wallet environment was connected to the tester's local RPC endpoint.

Source-of-Truth Priority

Priority	Source	Reason
1	Local RPC + Explorer	Used to verify actual on-chain state when public RPC was stale or failing.
2	Transaction receipts & raw traces	Used to confirm contract execution, value movement, internal transfers, and burn mechanics.
3	Backend API responses	Used to inspect profile, history, queue, recovery, badge mapping, and leaderboard state.
4	Frontend UI	Used to validate what users actually see and where misrepresentation occurs.

Consolidated Severity Matrix

Area	Confirmed Issue	Severity	Primary Risk
Public RPC / Network State	Public RPC lagged thousands of blocks behind local RPC; returned null receipts for confirmed local transactions; produced 502/504/connection reset errors.	High	Frontend confirmation, balance display, recovery workflows, and indexer behavior can become unreliable.
Faucet	Claim cooldown can start when delivery is still pending and no tx hash exists; queue state can reference old entries.	High	Users lose daily claim opportunity without confirmed DACC delivery.
QE Pool Burn	1 DACC burn succeeded on-chain, but +1000 QE was not credited and burn tx was absent from exchange/history after recovery polling.	High	Real loss-of-credit risk: DACC spent/burned on-chain while reward accounting remains unchanged.
Badge NFT State	Vanguard and Sentinel Rank NFTs were minted on-chain, but profile.nft_tx_hash remained empty and UI showed MINT NFT.	High	Users can be misled into duplicate mint attempts and gas loss.
Settings / Social Binding	Linked state is visible, but Discord role eligibility, tier reasoning, refresh/rebind, and X account replacement are not exposed.	Medium-High	Users cannot self-correct stale identity/role state.
Leaderboard	Metadata reports zero/empty stats while leaderboard rows exist; search/sort/pagination operate only on loaded subset.	Medium	Users misinterpret partial dataset as global ranking.
Quantum Crate	QE reward accounting worked, but UI/history did not always update immediately; DACC drop pending state lacks queue/status transparency.	Medium	Users cannot confirm DACC reward settlement and may see stale post-open state.
My Activity	TX count and DACC balance appear sourced from different states; sync response is not applied atomically.	Medium	Activity page can be internally inconsistent.

Cross-System Pattern: State Desynchronization

The strongest recurring pattern is not a single isolated bug. It is a systemic consistency problem across multiple state layers. When any middle layer lags, fails, or returns stale data, the UI can continue presenting outdated information as if it were current. In modules where user funds or rewards are involved, this creates practical risk.

```
Wallet / Local RPC / Explorer state
  ↓
Public RPC / Indexer / Backend processing
  ↓
Profile API / History APIs / Queue APIs
  ↓
Frontend UI rendering and action state
```

State Boundaries Observed

Boundary	Example Observation	Impact
On-chain ↔ Public RPC	Public RPC stayed at block 14,912,620 while local RPC was above 14,919,000 during QE Pool testing.	Public RPC returned null receipts for confirmed transactions.
Public RPC ↔ Frontend	UI stuck at WAITING FOR BLOCK after explorer/local RPC confirmed tx success.	User cannot distinguish failed transaction from stale confirmation watcher.
On-chain ↔ Backend Profile	Burn tx succeeded on-chain but profile.qe_balance remained unchanged.	Reward accounting failed to reflect confirmed burn.
On-chain NFT ↔ Badge UI	Rank Vanguard/Sentinel minted on-chain but profile.nft_tx_hash empty and UI showed MINT NFT.	Duplicate mint risk.
Session ↔ Frontend state	Frontend still showed user as connected while protected endpoints returned 401.	Protected actions fail silently or confusingly.

Module Findings

6.1 Wallet, Session, Faucet, and Public RPC Baseline

The initial baseline established that a clean Incognito session reduced multi-wallet provider conflicts. In the main browser profile, several wallet extensions caused provider injection errors such as "Cannot redefine property: ethereum" and "Failed to set window.ethereum". This is likely an environment-level multi-wallet provider issue, although the web app could handle it more gracefully.

Authentication and Session Behavior

Before login, /api/inception/profile/ returned 401 Unauthorized. After successful wallet login, the same endpoint returned 200 OK. This behavior is expected. However, later during v0.4 testing, the frontend remained visually logged in while authenticated endpoints returned 401 Not authenticated, indicating session desynchronization.

Public RPC Balance Inconsistency

```
Local RPC block : 0xe39bc6 = 14,916,550
Local balance   : 3.990335636 DACC

Public RPC block : 0xe38a39 = 14,912,057
Public balance   : 5.095386530 DACC

Block gap       : 4,493 blocks
```

DAC Inception's top bar matched the public RPC balance rather than the synced local RPC/explorer balance. A same-block hash check did not prove a fork; the stronger conclusion is public RPC lag.

Faucet Claim Flow

```
faucet/ response after claim:
{ "success": true, "status": "pending", "dispense_id": "e232ee2b-b9cf-4c9e-8b7a-e71a6832477c"
}

status for dispense ID:
{ "status": "pending", "tx_hash": "", "error": "" }
```

The faucet request was accepted, but no confirmed delivery or tx hash was returned. The UI still started a 24-hour cooldown. After restart/relogin, cooldown persisted while local RPC/wallet balance did not show a +3 DACC increase.

Faucet Queue State

```
{ "in_queue": true, "total_pending": 5559466, "status": "pending",  
  "ahead_of_you": 3889219, "submitted_at": "2026-05-08T05:22:40.797800+00:00" }
```

The queue entry referenced an old May 8 timestamp and did not clearly correspond to the latest dispense_id. Faucet history appeared to show global recent claims, not per-wallet delivery proof.

6.2 My Activity State Consistency

My Activity appears to mix multiple sources: profile/ for QE, rank, streak, tx count, badges, and metadata; qe-history/ for earned QE; public RPC for DACC balance; sync/ for manual task recalculation; and network/ for live indicators.

Sync Non-Atomicity

```
Before sync: dacc_balance = 7.951507 | tx_count = 964  
Sync response: dacc_balance = 6.079370 | tx_count = 983  
Profile after: dacc_balance = 3.990336 | tx_count = 1006
```

The UI reflected the updated TX count but continued showing DACC balance tied to the public RPC/top bar value. One panel can display values sourced from different state layers.

TX Count Ambiguity

Observed values: UI/profile TX 964 → sync response 983 → post-sync profile/UI 1006 → explorer showing 1,169 transactions. This may be intended if the UI is showing indexed or qualified Inception transactions, but the label does not explain that distinction.

6.3 Leaderboard Data and Global Ranking Behavior

Leaderboard rows loaded, but metadata fields were zero or empty: total_count = 0, total_users = 0, level_distribution = []. The UI displayed "No operatives yet" in Level Distribution while actual ranked rows were visible.

Client-Side Search, Sort, and Pagination

- Searching for JERUZZALEM produced no backend request and cleared the table because the user was outside the top loaded subset.
- Sorting by QE, Badges, Tasks, Refs, and Streak did not trigger backend requests.
- Pagination to page 2, 3, 4, and beyond did not trigger backend requests.
- The UI gives the impression of global search/sort/pagination, but behavior indicates client-side operations over a limited preloaded dataset.

Rank Freshness

The tester's displayed personal rank remained #4649 for an extended period, even after intentionally breaking streak and skipping activity. The UI should disclose whether rank is real-time, periodically recalculated, or snapshot-based.

6.4 Settings, Social Binding, and Identity Sync

Settings showed the tested wallet as linked across core identity surfaces: email verified, X linked/following, Discord linked/joined, Telegram joined, and profile greeting as JERUZZALEM. Basic boolean fields matched profile/.

Role-Level Eligibility Not Transparent

Observed Discord roles: Waiting Room, Commando, Cadet lvl1, Verified. Profile fields: waitlist_qe: 0, early_badge_type: standard, discord_linked: true, discord_joined: true. The app confirms Discord linkage but does not expose role-level eligibility, role sync time, gold eligibility, clearance reason, or waitlist eligibility reason. Users cannot verify whether role-based benefits are synchronized.

No Rebind or Refresh Path

After accounts are linked, Settings does not expose refresh, re-check roles, reconnect, unlink, replace linked X account, or manually re-verify eligibility. This can trap users with stale or suspended social accounts.

6.5 Quantum Crate

Quantum Crate was tested through five daily opens. QE reward accounting worked correctly after state refresh, but immediate UI/history refresh lag and DACC-drop settlement opacity remain issues.

QE Reward Accounting

```
Starting QE : 33,562
Open #1      : +250 QE      Open #2 : Empty
Open #3      : +1500 QE     Open #4 : +100 QE
Open #5      : Empty

Total reward  : 1,850 QE
Total cost   : 750 QE      (5 × 150)
Net          : +1,100 QE
Expected final: 34,662     Profile final: 34,662 ✓
```

Backend/profile final state matched expected accounting. Daily limit reached 5/5 and qe_today reached 1,850/3,000.

UI/History Desynchronization

After the first open, open/ returned opens_today = 1 and qe_today = 250, but history/UI initially still showed opens_today = 0 and qe_today = 0. Manual in-page refresh fixed the state. The recommendation is to refetch history/profile after open/ success and avoid relying on manual user refresh.

DACC Drop Pending UX

```
reward_label : 1 DACC
reward_type   : dacc
dacc_tx_hash  : pending:09c17523-46d2-4a6b-b484-dd1cd06da52c
opened_at    : 2026-05-23T09:47:20.585154+00:00
```

This was not proven as an on-chain delivery failure because the original real-time debug was not captured. It is a confirmed UX issue: users who receive DACC drops do not have a clear queue/status view showing whether the drop is pending, queued, failed, retrying, or completed.

6.6 QE Pool / Exchange

QE Pool was tested through controlled active transactions: stake 1 DACC, stake 1 DACC again, unstake 1 DACC, and burn 1 DACC. The contract mechanics were mostly valid; the application-layer state was not.

Stake / Unstake Contract Behavior

- Stake #1 succeeded on-chain: 1 DACC sent to contract, internal fee settlement back to wallet of 0.78233075 DACC.
- Stake #2 succeeded on-chain: 1 DACC sent to contract, internal fee settlement of 0.000150518 DACC.
- Unstake succeeded on-chain: user sent 0 DACC call; contract returned 1 DACC principal plus 0.000846585 DACC fee reward through internal transactions.

Public RPC vs Local RPC Divergence

```
After 2 stakes + 1 unstake:

Local RPC  block   : 14,918,990
           balance: 3.773663478993 DACC
           stake   : 58 DACC
           pending: 0.000182521 DACC

Public RPC block   : 14,912,620
           balance: 5.095386530154 DACC
           stake   : 57 DACC
           pending: 0.721957388938 DACC
```

The UI continued showing stale Your Stake = 57 DACC and Pending Fees = 0.721957 DACC even though local/on-chain state showed 58 DACC and a much smaller pending fee.

Burn 1 DACC and Missing QE Credit

```
Burn TX: 0xledbdb7821d4cb97c1be073b1a92dcab46643cf22f00639455d7d1830517089d

On-chain result:
  User → Contract      : 1 DACC
  Contract → 0x000...dead: 0.95 DACC
  Remaining              : 0.05 DACC (LP fee)

Expected QE after burn : 35,662
Actual profile.qe_balance: 34,662
Delta                   : -1,000 QE (NOT CREDITED)
```

The burn contract mechanics matched the stated 95% burn / 5% LP fee model. The +1000 QE credit did not appear in profile, and exchange/history did not include the new burn transaction during the observation window.

Recovery Flow

```
recover-burns/ response: { "status": "queued", "next_check_in": 600 }

After >10 minutes:
  qe_balance      : 34,662 (expected: 35,662)
  burn_tx_in_history : false
  public_rpc_block  : 14,912,620
  public_rpc_receipt : null
  Additional errors  : 502 Bad Gateway, ERR_CONNECTION_RESET
```

The missing QE recovery mechanism exists and can be queued, but it did not resolve the confirmed burn within the observation window. This is the highest-severity reward-accounting issue in the report.

6.7 Badges / Rank NFT State

Badge validation confirmed 63 earned profile badges and a badge QE sum of 9,735. The issue is specifically incomplete Rank NFT transaction mapping: profile records six earned rank badges but only four nft_tx_hash values.

Profile vs On-Chain Rank NFT State

Rank	Profile Earned	Profile nft_tx_hash	On-chain Mint Proof
Cadet	Yes	Present	Confirmed
Commando	Yes	Present	Confirmed
Seal	Yes	Present	Confirmed
Shadow Unit	Yes	Present	Confirmed

Vanguard	Yes	Missing	Confirmed
Sentinel	Yes	Missing	Confirmed
Sovereign	No	N/A	Not claimed

Local RPC Mint Logs

rank_cadet	tx: 0xca598e1aa2398ebde70487336e87da061ce191af43fdb0fa54aff31e006bb973
rank_commando	tx: 0x9cdbf01cedec8179096d9e349ea6c3cc80a32e15eac803d60dad32d1221b6532
rank_seal	tx: 0x2b61c4408f8774ecd77b8b80c4039c1f0d11553af74ab41f7137db56573130f8
rank_shadow	tx: 0xd6636f10d02f1c4f197a64f0b30c682a46786cfe53b8823a5d836333152a35a
rank_vanguard	tx: 0x91a41be3deec3b65b0c975afe841a56566abef28e47f29bc43efea9e32f7b3aa
rank_sentinel	tx: 0x4a573d6990376042ef74e2bfca52a21c1c468819a53e7cb842b0579174c6c7b0

The DAC explorer inventory also showed six DAC Inception Rank NFTs held by the wallet. Vanguard and Sentinel are already minted on-chain even though the frontend shows MINT NFT.

Duplicate Mint Risk

A previous report (Report 4, 10 May 2026) documented the same class of issue: a successful NFT mint was confirmed on-chain but the frontend kept the mint button active for an extended period. A duplicate mint attempt was submitted, the contract rejected it, and gas was consumed. The current Vanguard/Sentinel state is consistent with that pattern.

Technical Analysis

Several smart contract paths performed correctly when checked through local RPC and explorer: QE Pool stake/unstake handled internal fee settlement correctly; burn sent 0.95 DACC to the dead address and allocated 0.05 DACC as fee share; Rank NFTs were minted to the wallet. The observed failures occur after that — confirmation watchers, backend profile, indexer history, queue status, or frontend UI do not consistently reflect confirmed state.

Root-Cause Class	Description	Affected Modules
Public RPC lag/failure	Public RPC can be behind local chain state or return 502/504/connection reset.	Balances, QE Pool, Badges, faucet confirmation, transaction watchers
Indexer/event listener delay	Backend/indexer may not process confirmed on-chain events into history/profile.	QE Pool burn history, Rank NFT nft_tx_hash mapping
Frontend stale-cache/state issue	UI uses cached or partial state and does not reconcile after successful actions.	Quantum Crate, My Activity, Badge UI, QE Pool waiting state
Backend session desynchronization	Frontend appears connected while authenticated endpoints return 401.	Settings, QE recovery, My Activity
Ambiguous data-model semantics	TX count, badge count, leaderboard scope, Total QE vs Testnet QE are not explained.	My Activity, Badges, Leaderboard

User Impact and Risk Assessment

- Users can be shown stale balances and incorrectly infer that faucet delivery, reward credit, or transaction state is correct or failed.
- Users can enter cooldown after faucet request acceptance without confirmed DACC delivery.

- Users can burn DACC on-chain while the expected QE remains uncredited in profile/history.
- Users can be shown MINT NFT for already-minted badges and may submit duplicate mint attempts that fail and consume gas.
- Users cannot self-diagnose Discord role eligibility, badge tier, waitlist QE, or social rebind problems.
- Leaderboard users may interpret client-side subset sorting/search as global ranking behavior.
- Community trust is impacted because the dashboard does not reliably reflect on-chain state or backend accounting state.

Recommendations and Remediation Roadmap

P0 — Immediate Reliability and Safety Fixes

1. Add RPC health detection and stale-state warnings directly in the UI. Display public RPC block, block age, and degraded/stale status.
2. Add fallback confirmation logic for transactions. If public RPC is stale, fall back to an indexed source, explorer-compatible backend, or user wallet/local RPC where feasible.
3. For burn-to-QE conversion, make reward crediting idempotent and recoverable. A confirmed burn should result in exactly one QE credit even if the first indexer pass fails.
4. Before showing MINT NFT, check whether the wallet already owns the corresponding Rank NFT. Empty `profile.nft_tx_hash` must not be treated as proof of unminted state.
5. When protected endpoints return 401, immediately clear stale authenticated frontend state and prompt the user to reconnect.

P1 — User Transparency and Recovery

1. Add per-user faucet claim status with claim ID, queue position, dispense ID, tx hash, status, failure reason, and cooldown reason.
2. Add QE Pool pending/uncredited burns list with statuses: on-chain confirmed, indexed, QE credited, recovery pending, failed.
3. Make recover-burns/ return scan result details: scanned block range, found count, recovered count, skipped count, and failure reason.
4. Add Rank NFT re-sync action allowing users to refresh already-owned NFT state or submit a mint tx hash for reconciliation.
5. Add Settings re-check/rebind controls for Discord roles, X account changes, Telegram, and eligibility refresh.

P2 — Product Clarity and Observability

1. Clarify TX count semantics: total explorer transactions vs indexed/qualified Inception transactions.
2. Clarify leaderboard scope: global server-side search/sort/pagination vs currently loaded subset.
3. Add last-updated timestamps to leaderboard rank, profile, badge state, QE Pool state, and social eligibility state.
4. Clarify Total QE vs Testnet QE and what variables influence Leaderboard rank.
5. Add monitoring for indexer lag, missed events, queue aging, stuck DACC drops, and reward-credit mismatches.

Developer Validation Checklist

Check	Expected Result
Compare public RPC head vs reference node head	UI shows degraded/stale status if block gap exceeds threshold.
Fetch wallet balance from public and local/reference RPC	Discrepancy is detected and not silently displayed as final truth.
Submit faucet claim and track dispense ID	Cooldown starts only with documented pending/delivery state; tx hash appears when submitted.
Run recover-burns after a known confirmed burn	Recovered burn appears in exchange/history and profile.qe_balance increments exactly once.
Check Rank NFT contract logs for wallet	profile.nft_tx_hash maps all already-minted rank NFTs.
Attempt UI state after successful mint	Already-owned badge shows ON-CHAIN or Sync Pending, not MINT NFT.
Expire backend session while frontend remains open	Frontend detects 401 and prompts reconnect.
Leaderboard search for rank outside top loaded subset	Server-side search finds the user globally or UI labels search as local-only.

Conclusion

This investigation shows a repeated application-layer consistency problem in DAC Inception Testnet Web v0.9.7. The blockchain layer and several contract operations work correctly, but frontend/backend/indexer/public RPC state may fail to reflect confirmed on-chain reality.

The highest-risk finding is reward accounting around QE Pool burns: a 1 DACC burn was confirmed on-chain, but the expected +1000 QE remained uncredited in profile and absent from exchange/history after recovery was triggered and observed for more than 10 minutes. The second highest-risk finding is badge NFT state: Vanguard and Sentinel are minted on-chain but shown as mintable because backend/profile nft_tx_hash mapping is incomplete.

The recommended priority is to improve RPC/indexer reliability, introduce explicit stale-state warnings, make reward and NFT synchronization idempotent and recoverable, and provide user-facing recovery and status tools for faucet, QE Pool, crate DACC drops, and Rank NFT state.

Final Engineering Summary: The repeated pattern is not "user action failed." In multiple cases, user action succeeded on-chain. The failure is state propagation: on-chain truth does not reliably reach the backend profile, indexer history, reward accounting, or UI rendering layer.

Appendix A — Evidence Index

Evidence Area	Key Evidence
Public RPC lag	Public RPC block 14,912,620 while local RPC was at 14,919,000+ during QE Pool tests.
Faucet pending	dispense_id e232ee2b...; status: pending; tx_hash: empty; cooldown: active.

Queue stale	Queue submitted_at remained 2026-05-08 while latest faucet claim had a new dispense ID.
Quantum Crate accounting	33,562 QE → 5 opens → 34,662 QE; opens_today: 5; qe_today: 1,850.
Old crate DACC pending	pending:09c17523-46d2-4a6b-b484-dd1cd06da52c
QE Pool stake #1	0xf97cae4df43c0d819f4f08c840b41bd45ccc2a5b10b84ae67e06690de9177fe3
QE Pool stake #2	0xd8803afc6c2615c3dde4818c1ff4becb807856485a9e827f1b8994e3eeac97a5
QE Pool unstake	0xf7badff46ae6764cc3e5d4933332fd0ecc6bfff97ee70c79da788d7b8458d12
QE Pool burn — missing QE	0x1edbdb7821d4cb97c1be073b1a92dcab46643cf22f00639455d7d1830517089d — expected +1000 QE not credited.
Rank NFT Vanguard	0x91a41be3deec3b65b0c975afe841a56566abef28e47f29bc43efea9e32f7b3aa — profile.nft_tx_hash empty.
Rank NFT Sentinel	0x4a573d6990376042ef74e2bfca52a21c1c468819a53e7cb842b0579174c6c7b0 — profile.nft_tx_hash empty.

Appendix B — Checkpoint Coverage Map

Checkpoint	Included in Final Report Section
v0.1	Wallet/session baseline, faucet, public RPC, balance inconsistency, initial badge issue context
v0.2	My Activity state-source mixing, sync non-atomicity, TX count ambiguity
v0.3	Leaderboard metadata, local search/sort/pagination, rank freshness
v0.4	Settings/social binding, auth session desync, faucet finalized claim flow, QE Pool baseline/recovery
v0.5	Quantum Crate reward accounting, UI/history lag, DACC drop pending UX
v0.6	QE Pool stake/unstake/burn, public RPC divergence, missing QE recovery
v0.7	Badges/Rank NFT mapping, local RPC NFT mint logs, frontend mint-state desync